

t2g square lattice tight-binding derivation

Anderson Steckler

Nearest Neighbor

	$l = 1, m = 0, n = 0$	$l = 0, m = 1, n = 0$	$l = 0, m = 0, n = 1$
$E_{xy,xy}$	V_π	V_π	V_δ
$E_{xy,yz}$	0	0	0
$E_{xy,zx}$	0	0	0

This is for hopping in the x and y directions. The integrals all have the same sign, hence the Hamiltonian becomes. Using the Fourier transformation $c_i = \frac{1}{\sqrt{N}} \sum_k e^{i\mathbf{k} \cdot \mathbf{R}_i} c_k$

$$\begin{aligned}
 H &= t \sum_{R,\delta} c_{i+\delta}^\dagger c_i \\
 &= \frac{t}{N} \sum_{R,\delta} \sum_{q,k} e^{-i\mathbf{q} \cdot (\mathbf{R}_i + \delta)} e^{i\mathbf{k} \cdot \mathbf{R}_i} c_q^\dagger c_k \\
 &= \frac{t}{N} \sum_{\delta} \sum_{q,k} \delta_{qk} e^{-i\mathbf{q} \cdot \delta} c_q^\dagger c_k \\
 &= t \sum_{\delta} \sum_k e^{-i\mathbf{q} \cdot \delta} c_k^\dagger c_k \\
 &= t \sum_k \left[e^{-ik_x a} + e^{-ik_y a} + e^{ik_x a} + e^{ik_y a} \right] c_k^\dagger c_k \\
 &= 2t \sum_k [\cos(k_x a) + \cos(k_y a)] c_k^\dagger c_k
 \end{aligned} \tag{1}$$

Next Nearest Neighbor

$\xi = \pm 1$ and $\xi' = \pm 1$

	$l = \xi/\sqrt{2}, m = \xi'/\sqrt{2}, n = 0$	$l = 0, m = \xi/\sqrt{2}, n = \xi'/\sqrt{2}$	$l = \xi'/\sqrt{2}, m = 0, n = \xi/\sqrt{2}$
$E_{xy,xy}$	$\frac{3}{4}V'_\sigma + \frac{1}{4}V'_\delta$	$\frac{1}{2}V'_\pi + \frac{1}{2}V'_\delta$	$\frac{1}{2}V'_\pi + \frac{1}{2}V'_\delta$
$E_{xy,yz}$	0	0	$\frac{\xi\xi'}{4}(V'_\pi - V'_\delta)$
$E_{xy,zx}$	0	$\frac{\xi\xi'}{4}(V'_\pi - V'_\delta)$	0

The diagonal terms do not depend on the sign so we get.

$$\begin{aligned}
H &= t \sum_{\delta} \sum_k e^{-ik \cdot \delta} c_k^{\dagger} c_k \\
&= t \sum_k \left[e^{-ia(k_x+k_y)} + e^{-ia(k_x-k_y)} + e^{-ia(-k_x+k_y)} + e^{-ia(-k_x-k_y)} \right] c_k^{\dagger} c_k \\
&= t \sum_k [2 \cos(k_x a) + 2 \cos(k_y a)] c_k^{\dagger} c_k \\
&= 4t \sum_k \cos(k_x a) \cos(k_y a) c_k^{\dagger} c_k
\end{aligned} \tag{2}$$

for the off diagonal terms we get a minus sign for $a(-\hat{x}, \hat{y})$ and $a(\hat{x}, -\hat{y})$.

$$\begin{aligned}
H &= t \sum_{\delta} \sum_k e^{-ik \cdot \delta} c_k^{\dagger} c_k \\
&= t \sum_k \left[e^{-ia(k_x+k_y)} - e^{-ia(k_x-k_y)} - e^{-ia(-k_x+k_y)} + e^{-ia(-k_x-k_y)} \right] c_k^{\dagger} c_k \\
&= t \sum_k \left[(e^{-iak_x} - e^{iak_x})(e^{-iak_y} - e^{iak_y}) \right] c_k^{\dagger} c_k \\
&= t \sum_k [(2i \sin(k_x a))(2i \sin(k_y a))] c_k^{\dagger} c_k \\
&= -4t \sum_k \sin(k_x a) \sin(k_y a) c_k^{\dagger} c_k
\end{aligned} \tag{3}$$

Single Layer Hamiltonian

Our matrix elements can be represented by the following functions

$$\begin{aligned}
\varepsilon_{xy} &= -2t_{\pi} [\cos(k_x a) + \cos(k_y a)] - 4t_{2xy} \cos(k_x a) \cos(k_y a) \\
\varepsilon_{yz} &= -2t_{\pi} \cos(k_x a) - 2t_{\delta} \cos(k_y a) - 4t_{2yzzx} \cos(k_x a) \cos(k_y a) \\
\varepsilon_{zx} &= -2t_{\delta} \cos(k_x a) - 2t_{\pi} \cos(k_y a) - 4t_{2yzzx} \cos(k_x a) \cos(k_y a) \\
g &= +4t_g \sin(k_x a) \sin(k_y a)
\end{aligned} \tag{4}$$

and the matrix is in our basis of yz, xz, xy

$$H_0 = \begin{pmatrix} \varepsilon_{yz} & g & 0 \\ g & \varepsilon_{xz} & 0 \\ 0 & 0 & \varepsilon_{xy} \end{pmatrix} \tag{5}$$

Interlayer Hoppings

Label the interlayer integrals as U .

	$l = 0, m = 0, n = 1$	$l = 1, m = 0, n = 0$	$l = 0, m = 1, n = 0$
$E_{xy,xy}$	U_{δ}	U_{π}	U_{π}
$E_{xy,yz}$	0	0	0
$E_{xy,zx}$	0	0	0

our interlayer hopping hamiltonian is then in yz, xz, xy

$$H_{\perp} = \begin{pmatrix} t_{\perp} & 0 & 0 \\ 0 & t_{\perp} & 0 \\ 0 & 0 & t_{\perp xy} \end{pmatrix} \quad (6)$$